AWS VPC PEERING

Internship Project Report

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Project report covering VPC architecture, peering, routing, security, validation, troubleshooting and cleanup.

1. Abstract

Amazon Virtual Private Cloud (VPC) provides logically isolated networking for AWS resources. VPC peering allows resources in two VPCs to communicate using private IP addresses when the peering connection, route tables, and security rules are configured correctly. This project presents a design for connecting two VPCs with non-overlapping CIDR ranges. The design includes public and private subnets, a bastion host for administrative access, route-table configuration, security-group restrictions, a validation procedure, and cleanup steps.

2. Objectives

• Understand the purpose of VPC peering.

• Design two VPCs with non-overlapping CIDR ranges.

• Place a bastion host in a public subnet and application instances in private subnets.

• Configure routes for communication in both directions.

• Restrict SSH and ICMP traffic with security groups.

• Define tests to validate connectivity and troubleshoot common issues.

• Identify cleanup steps required after a lab.

3. Network Architecture

The design uses VPC A (10.10.0.0/16) and VPC B (10.20.0.0/16). VPC A contains a public subnet (10.10.1.0/24) for the bastion host and a private subnet (10.10.2.0/24) for App-A. VPC B contains a private subnet (10.20.1.0/24) for App-B. A VPC peering connection links the two VPCs. App-A and App-B are designed to use private IP addresses without public IPv4 addresses.

The editable architecture diagram source is included in the diagrams folder.

4. Resource and IP Address Plan

Resource	Planned CIDR / address	Purpose
VPC A	10.10.0.0/16	First network
Public subnet A	10.10.1.0/24	Bastion subnet
Private subnet A	10.10.2.0/24	App-A subnet
VPC B	10.20.0.0/16	Second network
Private subnet B	10.20.1.0/24	App-B subnet
Bastion host	10.10.1.10 + public IPv4	Administrative access
App-A	10.10.2.10	Private instance
App-B	10.20.1.10	Private instance

These are planned values. Actual assigned addresses may differ depending on the lab environment.

5. VPC Peering and Routing

A peering connection must be created between VPC A and VPC B and accepted before it can carry traffic. The route tables need routes to the remote VPC CIDR: VPC A's public route table sends 10.20.0.0/16 through peering and 0.0.0.0/0 through the internet gateway; VPC A's private route table sends 10.20.0.0/16 through peering; VPC B's private route table sends 10.10.0.0/16 through peering. Local routes remain in place. Both directions require routes so reply traffic can return.

6. Security Group Configuration

• Allow SSH to the bastion only from the user's current public IPv4 address using a /32 source.

• Allow SSH to App-A and App-B only from the bastion's private address (10.10.1.10/32).

• Allow ICMP Echo Request between the two private application instances only where needed for validation.

• Do not expose SSH or ICMP to all IPv4 addresses (0.0.0.0/0).

7. Connectivity Validation Procedure

• Test a ping from App-A to App-B before peering routes and security rules are configured.

• Configure the peering connection, routes and required security rules.

• Test ping from App-A to App-B, then from App-B to App-A.

• Access a private instance through the bastion host.

• Temporarily remove the relevant ICMP rule, repeat the test, restore the rule and verify recovery.

• Check EC2 instance details to confirm App-A and App-B do not have public IPv4 addresses.

For ping testing, the planned private addresses are 10.10.2.10 and 10.20.1.10. Actual observations should come from the AWS environment where the lab is performed. A design diagram or written procedure is not itself evidence of a successful live deployment.

8. Troubleshooting

If ping fails, verify that the peering connection is active, route tables are associated with the correct subnets, remote CIDR routes point to the peering connection, and security groups allow the intended ICMP traffic. If SSH fails, check the bastion public IP, SSH source restriction, key pair, login name, routes and security rules. A peering connection alone does not automatically provide connectivity.

9. Learning Outcomes

The project design covers VPC and subnet planning, non-overlapping CIDR ranges, VPC peering, route-table configuration, least-privilege security groups, bastion-host access, connectivity troubleshooting, and AWS resource cleanup.

10. Resource Cleanup

After the lab and evidence collection are complete, terminate the lab EC2 instances and review any EBS volumes, snapshots and Elastic IP addresses created. Remove the peering connection and related routes, security groups, subnets, route tables, internet gateway and VPCs as appropriate. Confirm that no unnecessary billable resources remain.

11. Conclusion

This report presents a VPC peering design and a structured procedure for configuring and validating private communication between two VPCs. The design depends on an accepted peering connection, routes in both directions, and appropriately restricted security-group rules. The validation procedure can be used to assess connectivity in the AWS lab environment.